



Multi-Agent Systems

Report

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Coordination of 5-bodies

First-order dynamics

1) What is the corresponding Laplacian matrix?

For an undirected weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$, the adjacency matrix $A = [a_{ij}]$ is defined as: if $(i,j) \in \mathcal{E}$, then $a_{ij} = w_{ij} > 0$ otherwise, $a_{ij} = 0$. In the degree matrix $D = diag(d_1, \dots, d_5)$, $d_i = \sum_{j=1}^{5} a_{ij}$. The Laplacian matrix is defined as: $L = D - A$.

In addition, this project uses the unweighted case the weight of every edge is $w_{ij} = 1$, so A is a symmetric 0-1 matrix. The adjacency matrix and degree matrix are respectively:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix}, D = \text{diag}(3,4,1,3,3).$$

Therefore, the corresponding Laplacian matrix is:

$$L = D - A = \begin{bmatrix} 3 & -1 & 0 & -1 & -1 \\ -1 & 3 & -1 & 0 & -1 \\ 0 & -1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 2 & -1 \\ -1 & -1 & 0 & -1 & 3 \end{bmatrix}.$$

2) Consider a control law with diffusive couplings (on velocities), i.e.

$$F_i = -m_i K \sum_{j=1}^{5} w_{ij} (\dot{R}_i - \dot{R}_j),$$

With K = 1.

a) What are the eigenvalues and eigenvectors of the Laplacian matrix?

1. Eigenvalue Analysis

For the Laplacian matrix L used in the simulation, the eigenvalues calculated by MATLAB are:

$$\lambda(L \approx -0.0000, 0.8299, 2.6889, 4.0000, 4.4812).$$

If the graph is connected, L has only one zero eigenvalue, and it corresponds to the consensus subspace.

2. Eigenvector Matrix

The eigenvector matrix, denoted as

$$V = [u_1\ u_2\ u_3\ u_4\ u_5], LV = V\Lambda,$$

This matrix satisfies the equation, where the eigenvalue matrix Λ is:

$$\Lambda = \text{diag}(-0.0000, 0.8299, 2.6889, 4.0000, 4.4812).$$

Based on the data, the normalized eigenvectors are listed below:

$$u_1 = \begin{bmatrix} 0.4472 \\ 0.4472 \\ 0.4472 \\ 0.4472 \\ 0.4472 \end{bmatrix}, u_2 = \begin{bmatrix} 0.2560 \\ -0.1380 \\ -0.8115 \\ 0.4375 \\ 0.2560 \end{bmatrix}, u_3 = \begin{bmatrix} -0.2422 \\ -0.5362 \\ 0.3175 \\ 0.7031 \\ -0.2422 \end{bmatrix}, u_4 = \begin{bmatrix} 0.7071 \\ 0 \\ 0 \\ 0 \\ -0.7071 \end{bmatrix}, u_5 = \begin{bmatrix} 0.4193 \\ -0.7024 \\ 0.2018 \\ -0.3380 \\ 0.4193 \end{bmatrix}.$$

The most important observation lies in the first column:

$$u_1 = \frac{1}{\sqrt{5}} 1,1 = [1,1,1,1,1]^\top,$$

The eigenvector corresponding to the zero eigenvalue aligns with the direction of all ones vector. This verifies the theory that "the null space of the Laplacian matrix characterizes the consensus subspace." In conclusion, when the velocities of all nodes are equal, the diffusive coupling term in the system becomes zero.

b) Calculate the consensus point. Confirm the results in simulation.

In this step, I rewrote the dynamics and control law from Section 2.1 into the standard form for velocity consensus.

The first order model provided in the problem is:

$$m_i \ddot{R}_i = F_i,$$

The control law (velocity diffusive coupling) is given by:

$$F_i = -m_i K \sum_{j=1}^{5} w_{ij} (\dot{R}_i - \dot{R}_j).$$

By substituting the control law back into the dynamic equation, we noticed a key point: the mass term m_i cancels out directly. This results in the following equation:

$$\ddot{R}_i = -K \sum_{j=1}^{5} w_{ij} (\dot{R}_i - \dot{R}_j).$$

Define the velocity as $v_i := \dot{R}_i \in \mathbb{R}^3$. Then, the equation becomes:

$$\dot{v}_i = -K \sum_{j=1}^{5} w_{ij} \left(v_i - v_j\right)$$

To get a more compact form (the standard consensus structure), I stacked the velocities of all agents into a single vector $v = \mathrm{col}(v_1, \dots, v_5) \in \mathbb{R}^{15}$. The system can now be expressed as:

$$\dot{v} = -(K\, L \otimes I_3)\, v.$$

Because the graph I selected is undirected and connected, the Laplacian matrix L is symmetric and positive semi-definite. The zero eigenvalue corresponds to vector 1 and the remaining eigenvalues are positive. Therefore, the velocities will converge to the same constant vector (velocity consensus).

Calculating the Final Consensus Velocity:

For consensus algorithms on undirected graphs, there is an invariant property: the sum of the states of all nodes remains constant. Therefore, if the system reaches consensus, it must converge to the average of the initial values.

The state is the velocity v_i, so the consensus velocity is:

$$v^\star = \frac{1}{5} \sum_{i=1}^{5} v_i\,(0).$$

The initial velocities given in the problem are:

$$v_1(0) = [10, -20, 10], v_2(0) = [-10, 10, 30], v_3(0) = v_1(0), v_4(0) = v_2(0), v_5(0) = 0.$$

$$\sum_{i=1}^{5} v_i\,(0) = v_1 + v_2 + v_3 + v_4 + v_5 = [0, -20, 80],$$

Therefore

$$v^\star = \frac{1}{5}[0, -20, 80] = [0, -4, 16] \text{ m/s}.$$

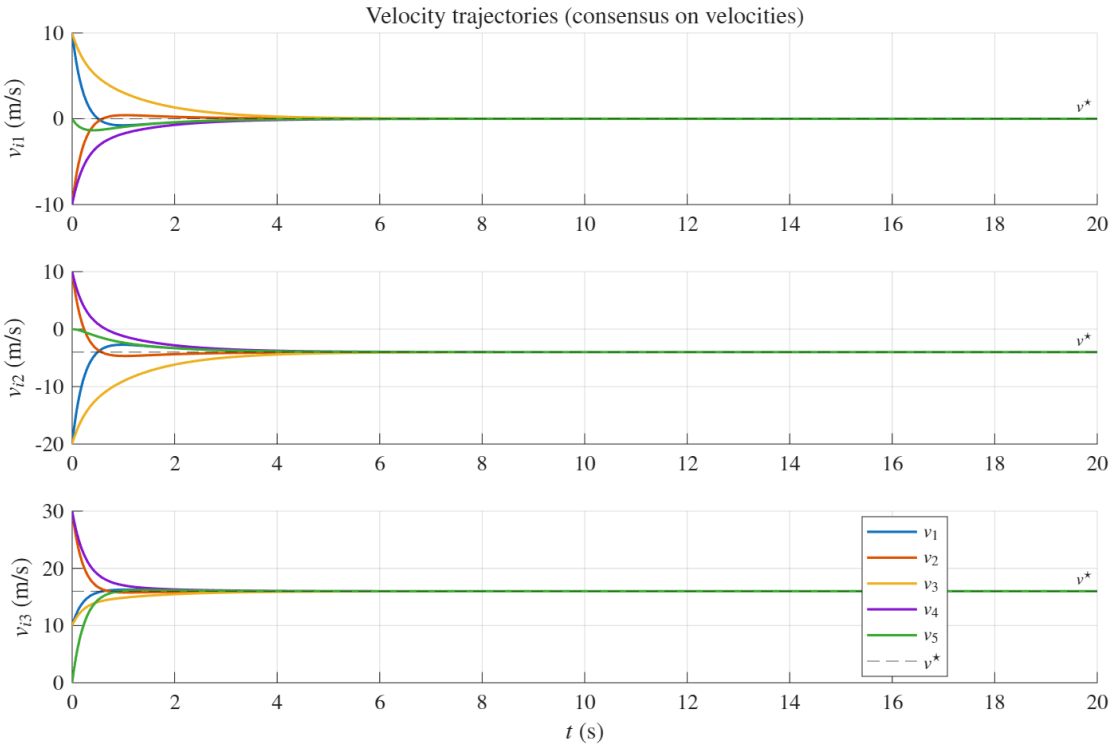


Figure shows the time evolution of the three velocity components of all five agents, the theoretical consensus value $v^\star$ is also plotted as a dashed line for comparison.

It can be observed that, despite different initial conditions, all velocity trajectories converge asymptotically to the same constant value along each spatial direction. The final value reached by all agents coincides with the predicted consensus velocity $v^\star$.

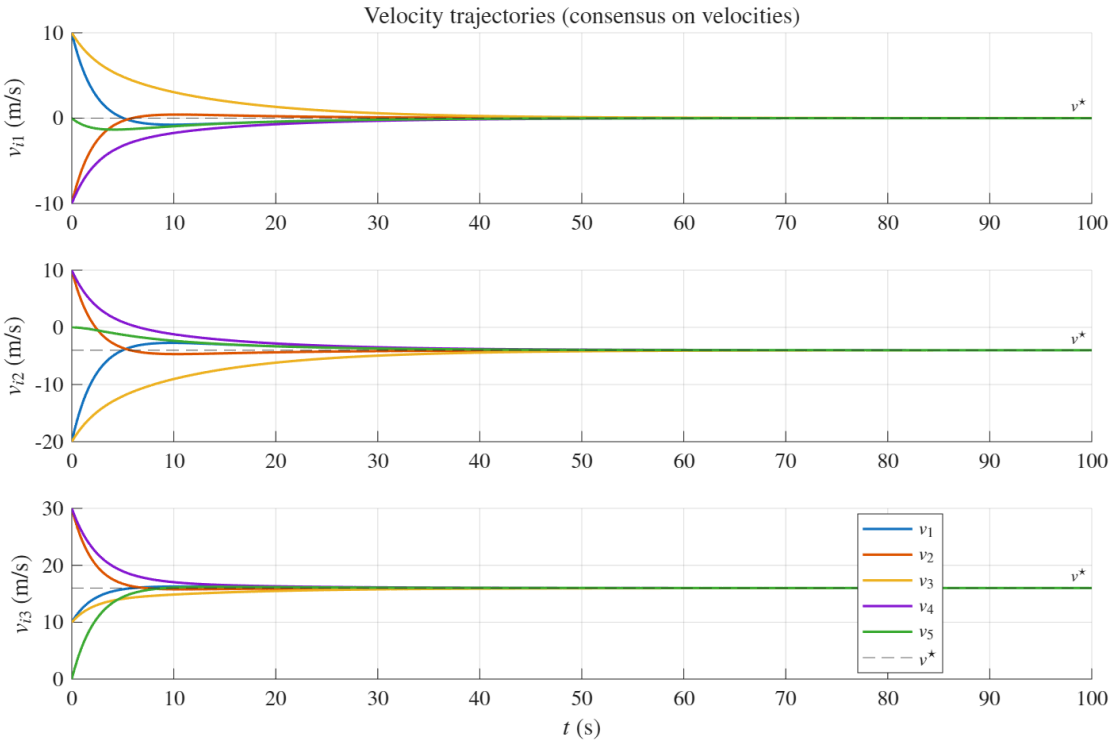
c) What do you observe in simulation when we choose the gain K as:

i) K = 0.1:

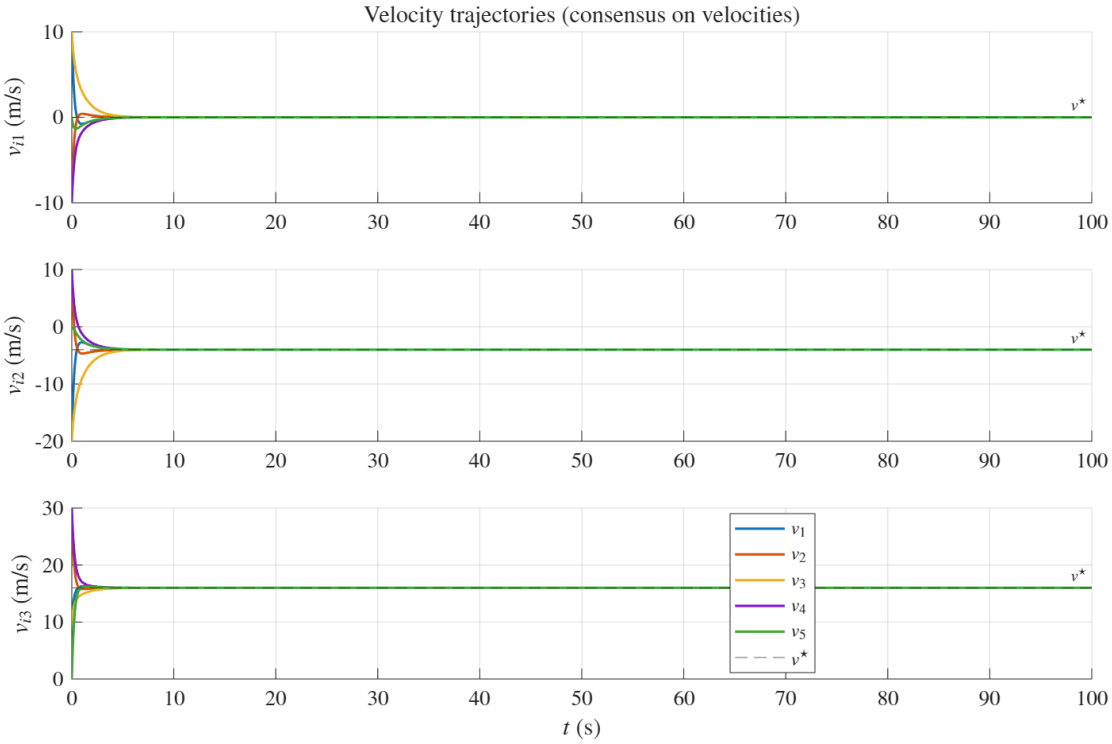
With a smaller control gain K = 0.1, it is observed from the simulation results that the velocities of all agents still eventually converge to the same constant value. This means that velocity consensus is still achieved. This indicates that, provided the communication graph is connected, reducing the control gain does not destroy the consensus property itself.

However, it can be clearly seen from the velocity trajectories that the convergence process becomes significantly slower. The velocity components take a longer time to gradually approach the consensus value, and the decay of the velocity error is very slow.

This phenomenon is consistent with the theoretical analysis: since the exponential decay rate of the non-zero modes in the closed-loop system is proportional to $K\lambda_2$, when K is small, the decay speed of the system's slowest mode is significantly reduced, resulting in an increased total convergence time.

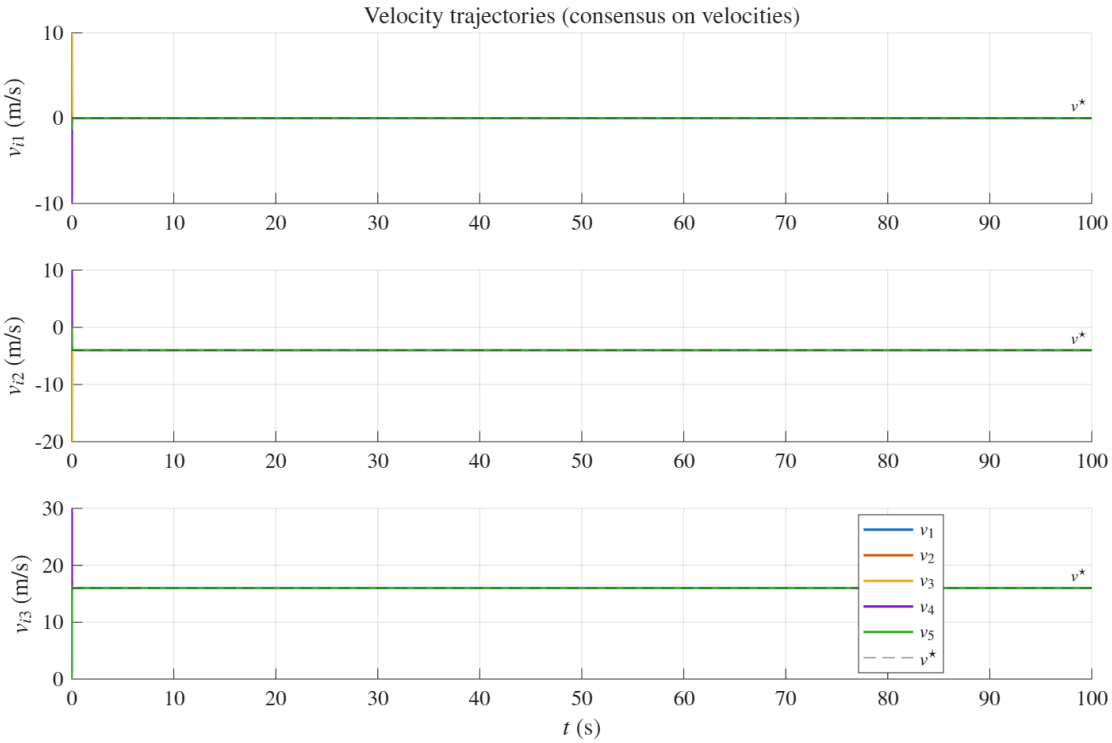


ii) K = 1:



For the case of K = 1, the system exhibits good convergence performance. It is observed from the simulation that the velocities of all agents converge rapidly to the same consensus value $v^\star$. within a short period. Furthermore, the trajectories are smooth, and no significant numerical oscillations are observed.

iii) K = 100:



When the gain is increased to K = 100, the convergence rate of the system is theoretically expected to increase significantly. This is because the decay rates of all non-zero modes are amplified by a factor of 100. This prediction is verified by the simulation results: the velocity trajectories are observed to converge to the consensus value $v^\star$ within an extremely short period, making the process appear almost instantaneous.

It is worth noting that, in this case, while the system remains theoretically stable and achieves velocity consensus, the system dynamics become extremely fast. Consequently, steep transient changes may be observed during the initial phase of the numerical simulation. This phenomenon does not indicate system instability; instead, it reflects that the high gain causes the system to become stiff, imposing higher demands on the numerical solver.

3) Consider the undirected version of the given graph topology. Reply to the following:

A) Measurement delays

We consider the undirected version of the given communication graph and study the effect of measurement delays on the consensus dynamics. The delay is assumed to be constant and uniform, $T_{ij} = 10\ \text{s}, \forall i, j$.

The distributed consensus control law on velocities is given by

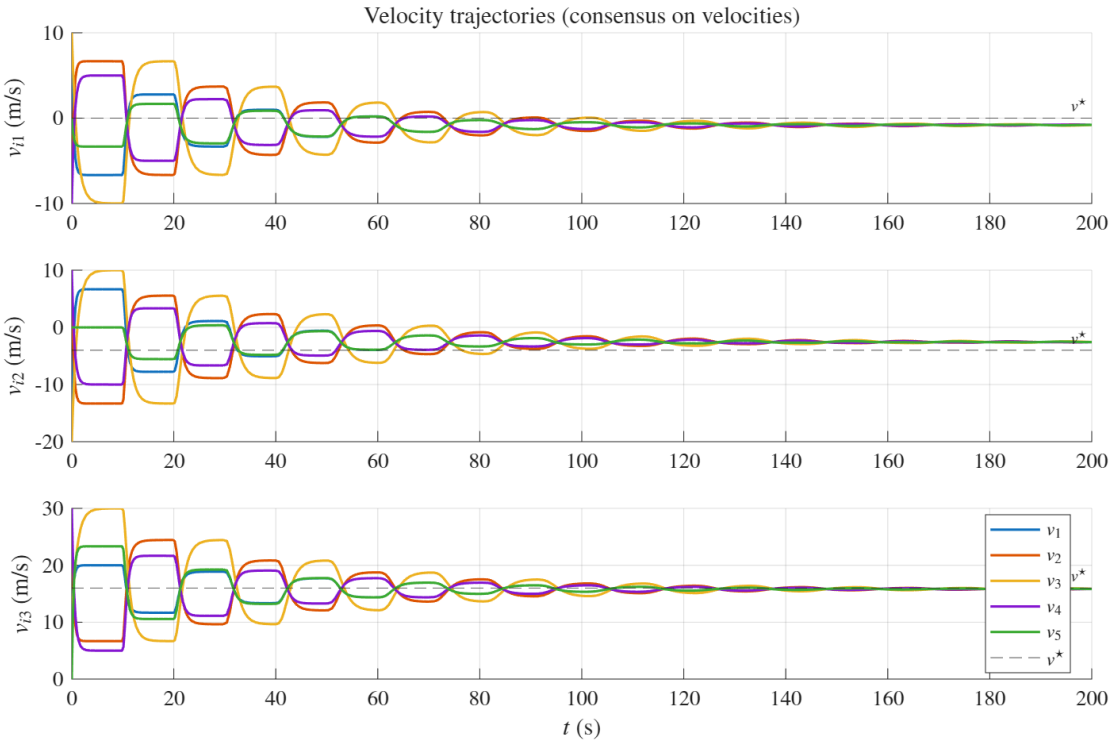
$$\dot{V}(t) = u_i(t) = -K\sum_j (v_i(t) - v_j(t - T_{ij})),$$

In matrix form, using the Laplacian structure introduced in class, this can be written as

$$\dot{V}(t) = -K(V(t)D - V_d(t)W),$$

From the course on consensus with time delays, we know that: For an undirected and connected graph, consensus can still be achieved in the presence of constant communication or measurement delays. However, delays reduce stability margins and typically introduce oscillatory transient behavior. Large delays slow down convergence and may cause oscillations, but do not necessarily

destroy consensus if the gain K is chosen appropriately.



The simulation results show that after introducing a 10s measurement delay, significant oscillations appear in the velocities of all agents during the initial phase (showing periodic changes). Subsequently, the oscillation amplitude gradually decreases, and all $v_i(t)$ eventually tend towards the vicinity of a common constant vector.

This indicates that under the current parameter settings, the system still possesses a tendency towards consensus. However, compared to the case without delay, the convergence time is significantly prolonged, and the process is accompanied by obvious oscillations.

Based on the readings from the curves, the criterion for convergence is defined as "all velocity components entering the vicinity of v* and remaining without significant deviation." It is estimated that the system enters the quasi-steady state region after approximately $t \approx 170 \sim 200\,\mathrm{s}$ (with slight differences in the time required for different components).

From the readings near the steady state, the final velocity values are observed to be very close to v*, although small deviations still exist. It is expected that if the simulation time is further extended and a more reasonable initial delay output is adopted, the trajectories will continue to approach v*. This would more clearly verify that delayed consensus can still be achieved under this value of K.

B) Input delay

In the case of input delay, the delay occur in the application of the control input to the system. The control input u(t) calculated at time t effectively enters the dynamics at time $t + T_{ip}$.

Consequently, for velocity consensus, the closed-loop system becomes a delay differential equation (DDE) with input delay. The typical form can be expressed as:

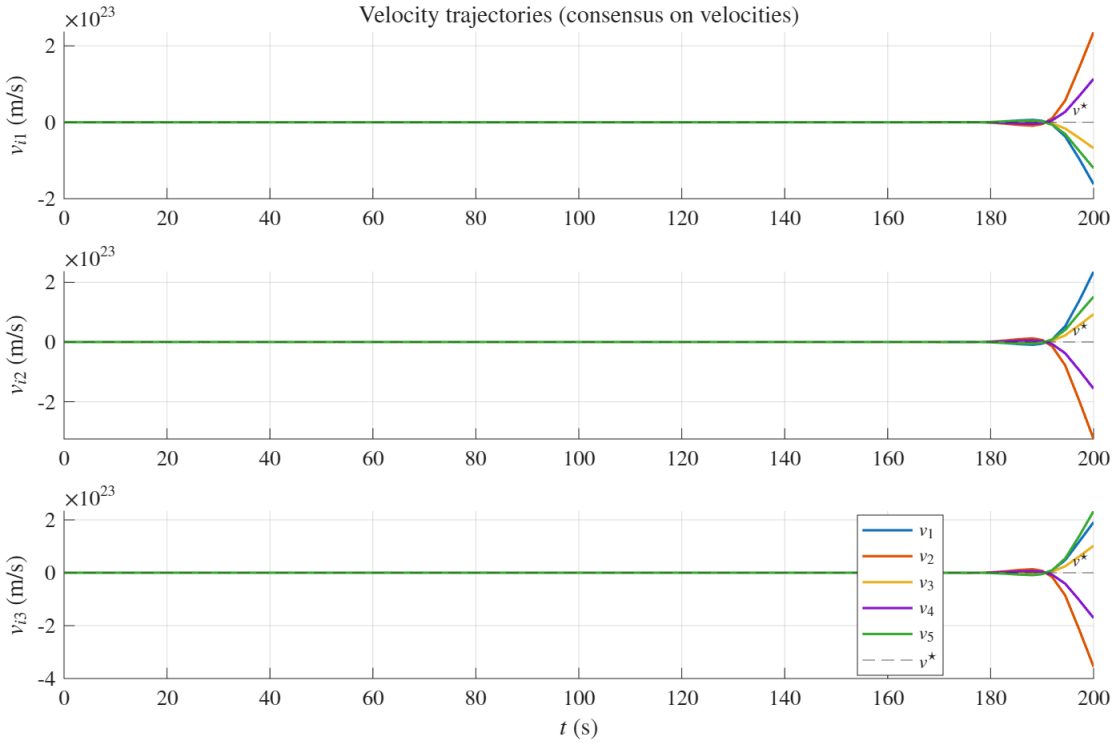
$$\dot{V}(t) = u(t - T_{ip}) = -KV(t - T_{ip})L$$

The theoretical implication is that, except for the consensus mode ($\lambda_1 = 0$), the error modes corresponding to all non-zero eigenvalues $\lambda_i(L) > 0$ satisfy a one-dimensional delay system.

Based on stability analysis, a commonly used sufficient condition for stability is derived. To ensure that all modes are stable, it is required that:

$$T_{ip} < \frac{\pi}{2K\,\lambda_{\max}(L)}$$

where $\lambda_{\max}(L)$ is the maximum eigenvalue of the Laplacian matrix.



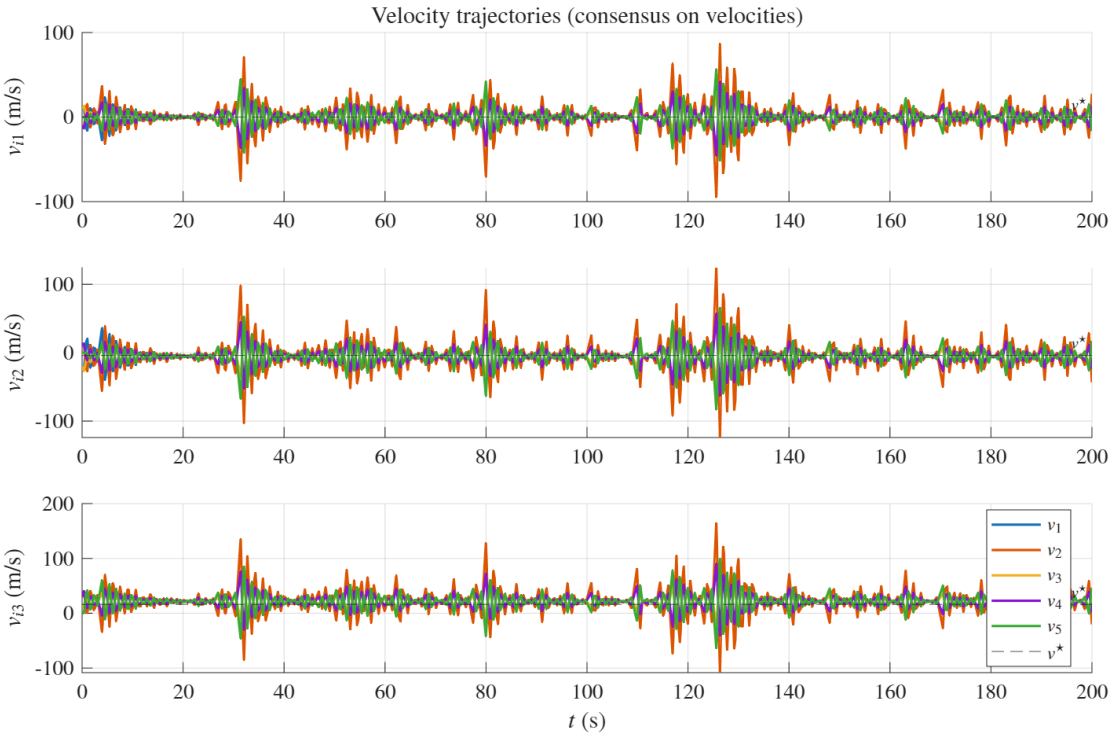
From the simulation results, it is observed that the velocity trajectories exhibit significant divergence in the final stage. The velocity values rapidly increase to a magnitude of 10^{23} (tending towards positive or negative infinity). This clearly indicates that the system has become unstable.

When $T_{ip} = 10$ s, consensus is not achieved. Instead, the system exhibits instability and divergence caused by the excessive time delay.

The maximum acceptable input delay that ensures consensus is:

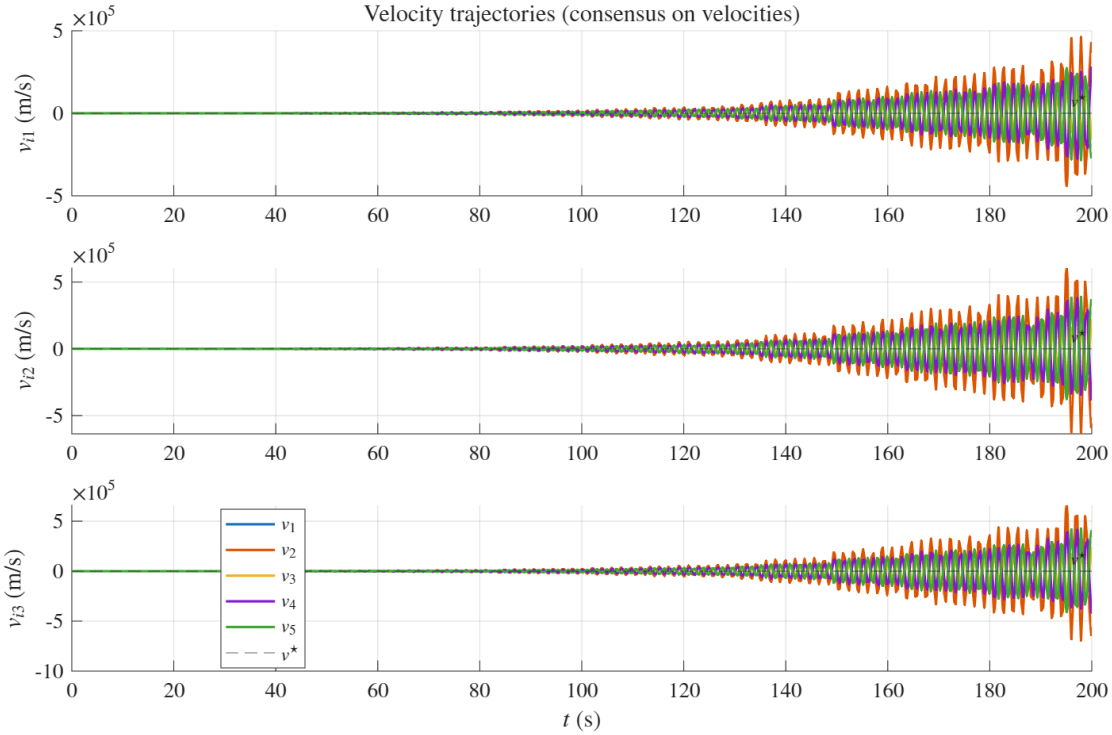
$$T_{ip}^{\max} = \frac{\pi}{2K\,\lambda_{\max}(L)} = 0.35s$$

In the simulation results, it is observed that the velocities do not converge smoothly to a constant line. Instead, persistent strong oscillations and spikes are present. These appear as high-frequency jitter (resembling noise), accompanied by occasional large-amplitude envelopes. This phenomenon is reasonable, because when the condition $T_{ip} = 0.35$ is met, the system operates near the boundary of critical stability. In this critical state, minor factors such as numerical errors, integration step sizes, and the initial function (since delay systems require a history period)—can easily drive the trajectories towards significant oscillation or even divergence. Therefore, $T_{ip} = 0.35$ should be regarded as a theoretical limit value. It is suggested that a smaller delay value be selected to obtain a clean and smooth convergence curve.



C) Measurement delays and Input delays

We consider the undirected communication graph affected by both constant measurement delays $T_{ij}$and constant input delays T_{ip}. In simulation, we set $T_{ij} = 10$ sand $T_{ip} = 10$ sfor all agents.



The simulation results show that consensus is not achieved. After an initial transient phase, the velocity trajectories start to exhibit oscillations whose amplitudes increase over time, eventually leading to divergence. The system becomes unstable and does not converge to the consensus value v^*.

This behavior can be explained by the accumulation of delays in the closed-loop dynamics. Measurement delays affect the information exchanged among agents, while input delays postpone the application of the control action. When both delays are present, the effective delay in the feedback loop becomes significantly larger, which severely degrades stability margins. As discussed in class, input delays are particularly critical, and the addition of measurement delays further reduces the maximum admissible delay ensuring consensus.

Therefore, although consensus may still be achieved when only one type of delay is present and sufficiently small, the simultaneous presence of large measurement and input delays leads to instability and loss of consensus, as confirmed by the simulation results.

2.2 Second-order dynamics

1.1) Choice of Gains Ki to Ensure Consensus

To ensure that the closed-loop stability can be analyzed using the standard results presented in the lecture (specifically the second-order consensus theory), the effect of different agent masses $m_i$must be addressed. If the ratio K_i/m_ivaries among agents, the system becomes a complex system coupled with non-uniform parameters, which is difficult to map to the standard Laplacian-based dynamics.

Therefore, to align the system with the standard form, the control gains K_i are chosen to be proportional to the mass of each agent:

$$K_i = m_i K, \quad K > 0.$$

By substituting this choice into the dynamic equation $m_i \ddot{R}_i = F_i$and dividing both sides by m_i, the mass term is effectively eliminated (cancelled out). Consequently, the closed-loop dynamics for both control laws are reduced to the standard forms characterized by the Laplacian matrix L and the scaling factor K.

After this normalization, the dynamics can be expressed in the standard matrix form (where $x_1 = R$ and $x_2 = \dot{R}$):

For Absolute Damping The system becomes:

$$\ddot{R}_i = -K \sum_{j=1}^{5} w_{ij} (R_i - R_j) - \gamma K \dot{R}_i$$

In state-space matrix form:

$$\dot{x}_2 = -(KL \otimes I_3)x_1 - (\gamma K I_5 \otimes I_3)x_2$$

For Relative Damping The system becomes:

$$\ddot{R}_i = -K \sum_{j=1}^{5} w_{ij} (R_i - R_j) - \gamma K \sum_{j=1}^{5} w_{ij} (\dot{R}_i - \dot{R}_j)$$

In state-space matrix form:

$$\dot{x}_2 = -(KL \otimes I_3)x_1 - (\gamma KL \otimes I_3)x_2$$

Under this configuration, since the communication graph is undirected and connected, consensus is guaranteed provided that $K > 0$and $\gamma > 0$.

1.2) What is the consensus value in both cases?

(a) Case with Absolute Damping

In the closed-loop system with absolute damping, the Laplacian term eliminates the position difference, but the damping term $-\gamma K\dot{R}_i$remains active. When projected onto the average direction, the dynamics of the average position $\bar{R}(t)$satisfy:

$$\ddot{\bar{R}}(t) = -\gamma K\dot{\bar{R}}(t).$$

This represents a first-order exponential decay. By integrating the velocity equation, the final position (as $t \to \infty$) is obtained. The agents converge to a stationary common point defined by:

$$R^* = \lim_{t\to\infty} R_i(t) = \bar{R}(0) + \frac{1}{\gamma K}\bar{V}(0), \text{for } i = 1, \dots, 5.$$

Simultaneously, the final velocity for all agents decays to zero:

$$\dot{R}_i(t) \to 0.$$

(b) Case with Relative Damping

In the case of relative damping, both the position and velocity coupling terms involve the Laplacian matrix. Since $1^T L = 0$, both terms vanish when projected onto the average direction. Consequently, the acceleration of the center of mass is zero:

$$\ddot{\bar{R}}(t) = 0.$$

This implies that the average velocity remains constant throughout the process:

$$\dot{\bar{R}}(t) = \bar{V}(0) = \frac{1}{5}\sum_{i=1}^{5} \dot{R}_i\,(0).$$

Therefore, the consensus value is not a static point but a time-varying trajectory. All agents converge to the following group trajectory:

$$R_i(t) \to R^*(t) = \bar{R}(0) + t\,\bar{V}(0), \text{for } i = 1, \dots, 5.$$

In this case, the final velocity of every agent converges to the initial average velocity $\bar{V}(0)$.

1.3) What are the closed-loop eigenvalues in the second case?

To determine the closed-loop eigenvalues, the system dynamics are decoupled based on the eigenvalues of the Laplacian matrix L, denoted as

$$0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_5.$$

For each spatial component, the system is equivalent to a set of independent second-order modes. Specifically, for the k-th mode corresponding to the eigenvalue λ_k, the dynamics satisfy the characteristic equation

$$s^2 + \gamma K\lambda_k s + K\lambda_k = 0.$$

The closed-loop eigenvalues are the roots of this quadratic polynomial and are given by the formula:

$$s_{k,\pm} = \frac{-\gamma K\lambda_k \pm \sqrt{(\gamma K\lambda_k)^2 - 4K\lambda_k}}{2}.$$

Two distinct cases are observed from this formula.

First, for the consensus mode ($k = 1$), since $\lambda_1 = 0$, substituting this value into the equation yields two zero eigenvalues,

$$s_{1,1} = 0 \text{and} s_{1,2} = 0.$$

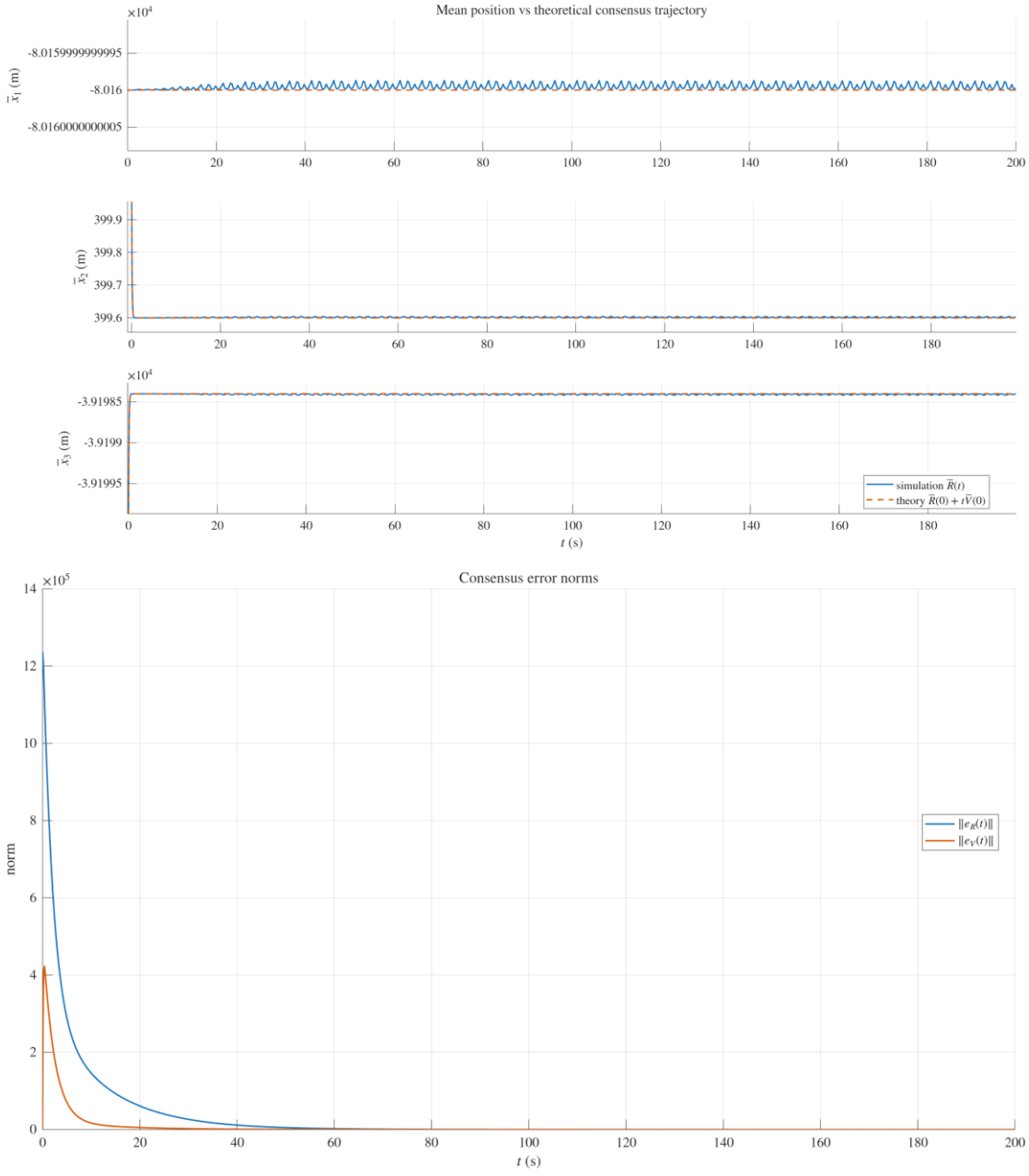
These roots correspond to the double-integrator dynamics of the average position, which results in a linear drift of the consensus trajectory as derived in the previous section.

For all remaining error modes ($k \geq 2$), the Laplacian eigenvalues are strictly positive ($\lambda_k > 0$). Given that the control gains $K$and $\gamma$are positive, the real parts of these eigenvalues are strictly

negative. This ensures that all disagreement errors decay to zero over time, thereby guaranteeing that consensus is successfully reached.

1.4) Verify in MATLAB/Simulink.

Absolute Damping:

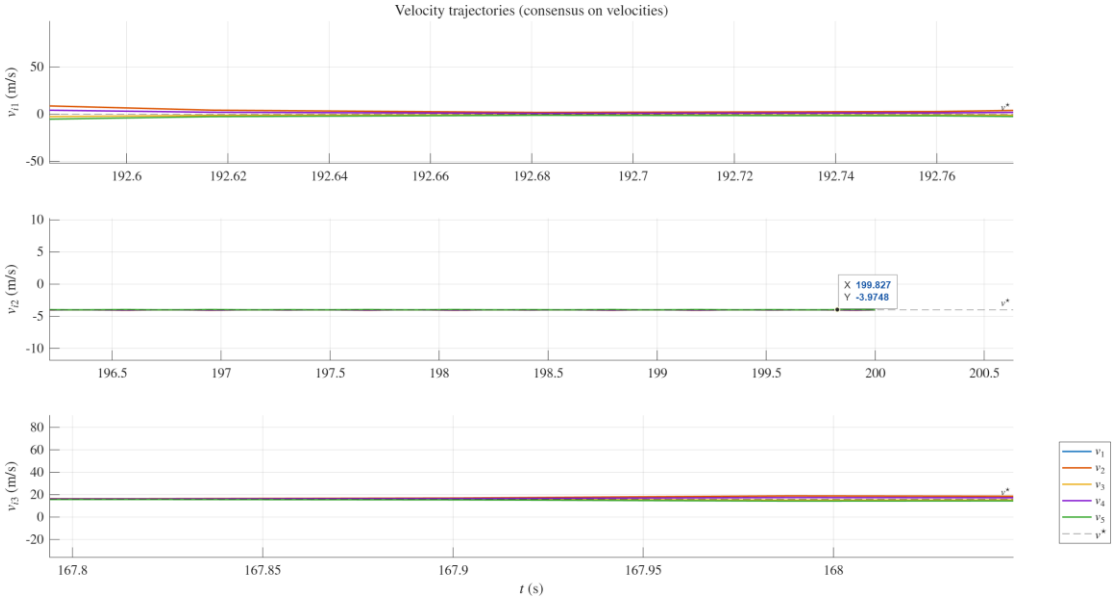


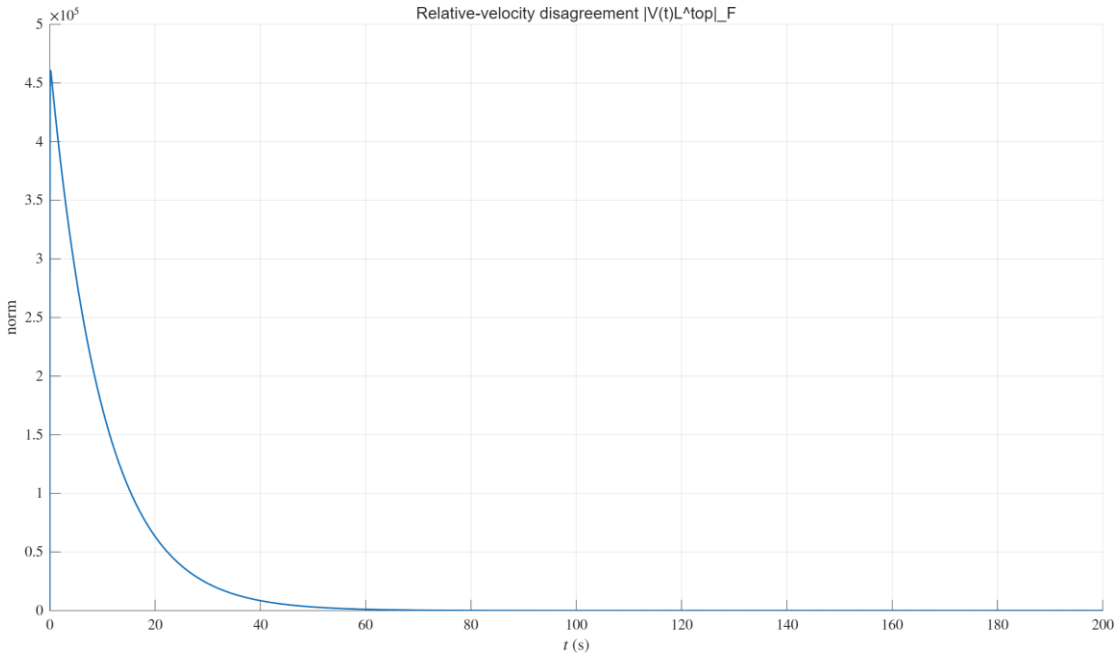
The simulation results regarding the consensus behavior of the system under the absolute damping control law are presented in the figure. From the comparison between the network average position and the theoretical trajectory, it is observed that the average position vector $\bar{R}(t)$rapidly converges to a constant value following a short transient phase. The system remains stable throughout the simulation process, with only minor high-frequency oscillations present. This phenomenon indicates that, under the influence of absolute damping, the overall motion of the system does not exhibit continuous drift but instead gradually ceases near a fixed location. This observation is consistent with the theoretical conclusions provided in the lecture.

Furthermore, based on the plot of the consensus error norms, it is evident that both the position error norm $\| e_R(t) \|$ and the velocity error norm $\| e_V(t) \|$ decay rapidly over time and eventually approach zero. Notably, the convergence speed of the velocity error is significantly faster than that of the position error. This suggests that the absolute damping term primarily suppresses the velocity differences among spacecraft first, and subsequently, the position offsets are eliminated through the position coupling term. Ultimately, both position and velocity consensus are achieved for all agents, with the velocity converging to zero as the system enters a static state.

In summary, the simulation results clearly verify the second-order consensus behavior under the absolute damping scenario. Not only is consensus achieved, but due to the damping acting directly on the absolute velocity, the overall velocity is dissipated, causing the network to finally converge to a stationary consensus position. This phenomenon is fully consistent with the analysis presented in the lecture slides regarding the property that “velocity vanishes while position converges”.

Relative damping:





The simulation results for the velocity trajectories indicate that the three velocity components $v_{i1}(t)$, $v_{i2}(t)$, and $v_{i3}(t)$for all agents converge to the horizontal dashed line representing v^*. This confirms that the final velocities of all spacecraft reach consensus and are consistent with the theoretically predicted average initial velocity.

To provide a more rigorous verification of network consensus, the relative-velocity disagreement function is plotted, defined as:

$$\eta(t) = \| V(t)L^T \|_F,$$
$$V(t) = [V_1(t), \dots, V_5(t)] \in \mathbb{R}^{3\times 5}$$

Since the condition $\eta(t) = 0$ holds if and only if

$$V_1(t) = \cdots = V_5(t),$$

meaning the velocities lie in the consensus subspace spanned by 1, the convergence of $\eta(t)$to zero directly indicates that the velocity differences among agents are eliminated.

It is observed from the simulation that $\eta(t)$decays rapidly to zero over time. This verifies that the closed-loop system with relative damping effectively ensures the decay of all non-zero Laplacian modes, thereby realizing velocity consensus. Combined with the result that the final velocity matches $v^* = \bar{V}(0)$, It is confirmed that the implemented control law behaves consistently with the theoretical conclusions.

2) Consider that all the masses m_i are identical

2.1) Selection of Gain Matrices $K_{1i}$and K_{2i}

Assumption and System Decomposition: In this analysis, it is assumed that all spacecraft have identical masses ($m_i = m$). Consequently, the entire network is modeled as a homogeneous second-order multi-agent system. This structure allows the stability analysis to be performed based on the eigenvalues of the Laplacian matrix L. Since the communication graph is connected, the Laplacian matrix Lhas exactly one zero eigenvalue (corresponding to the consensus state), while all other eigenvalues are strictly positive ($\lambda_k > 0$). The global dynamics can effectively be decoupled into independent modes corresponding to these eigenvalues.

Stability Requirement: For consensus to be ensured, it is required that the dynamic modes

corresponding to all non-zero Laplacian eigenvalues ($\lambda_k > 0$) are asymptotically stable. For each non-zero mode, the error dynamics degrades to a standard second-order system equation:

$$\ddot{z}_k + \lambda_k K_2 \dot{z}_k + \lambda_k K_1 z_k = 0.$$

To guarantee the convergence of this second-order system, the coefficients must be positively definite. This implies that the gain matrices $K_1$and K_2must be designed to ensure stability for all $\lambda_k > 0$.

Proposed Gain Selection: Based on the considerations above, isotropic diagonal matrices are selected for the gains: $K_{1i} = k_1 I_3$, $K_{2i} = k_2 I_3$, where $k_1 > 0$and $k_2 > 0$ are positive scalars.

This selection possesses clear physical interpretation: $k_1$represents the position coupling strength (determining the system stiffness), and k_2 represents the relative damping strength (used to suppress velocity differences). Under this configuration, since all relevant eigenvalues λ_k are positive, every non-zero mode behaves as a stable damped oscillator. Therefore, all relative position and velocity errors are guaranteed to decay to zero, ensuring that consensus is achieved.

2.2) Calculate the consensus trajectory. Verify in simulation.

To determine the consensus trajectory, the dynamics of the network's center of mass are analyzed. The average position $\bar{R}(t)$ and average velocity $\bar{V}(t)$ of the group are defined as:

$$\bar{R}(t) = \frac{1}{5}\sum_{i=1}^{5} R_i\,(t), \bar{V}(t) = \frac{1}{5}\sum_{i=1}^{5} \dot{R}_i\,(t).$$

Since the communication graph is undirected, the Laplacian matrix $L$is symmetric, and the sum of its rows is zero ($1^T L = 0$). Consequently, when summing the dynamic equations of all agents, the internal coupling forces (both position and velocity terms) cancel out efficiently. This leads to the following relationship for the average acceleration:

$$\ddot{\bar{R}}(t) = 0.$$

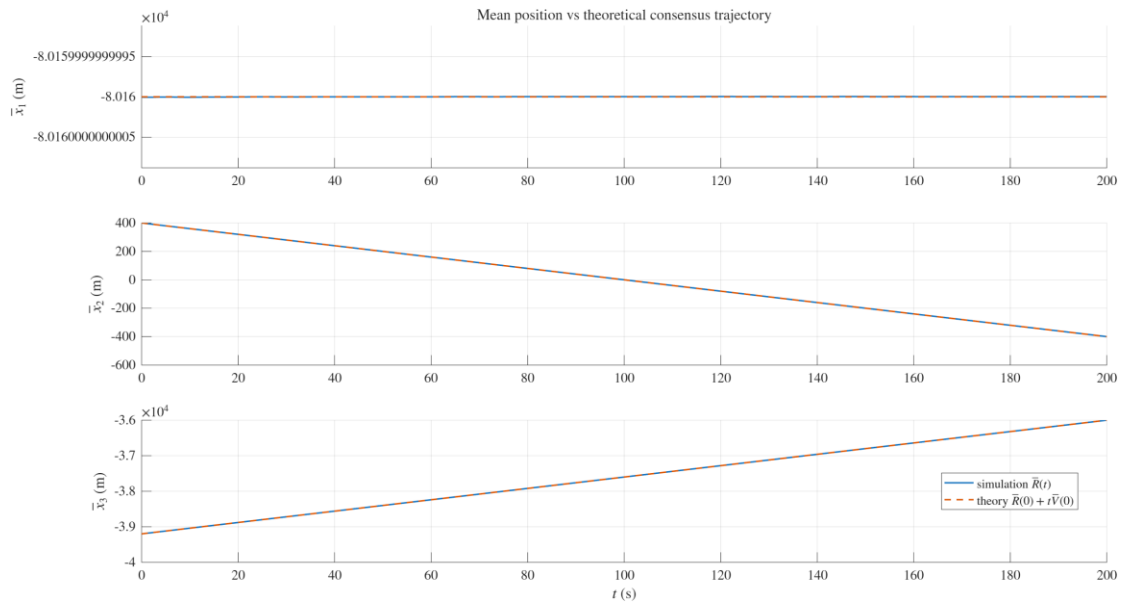
By integrating this equation, it is determined that the average velocity remains constant, equal to the initial average velocity:

$$\bar{V}(t) = \bar{V}(0).$$

Correspondingly, the average position follows a linear time-varying trajectory:

$$\bar{R}(t) = \bar{R}(0) + t\,\bar{V}(0).$$

Therefore, once consensus is achieved, the position of every agent $R_i(t)$is expected to converge to this specific linear trajectory.



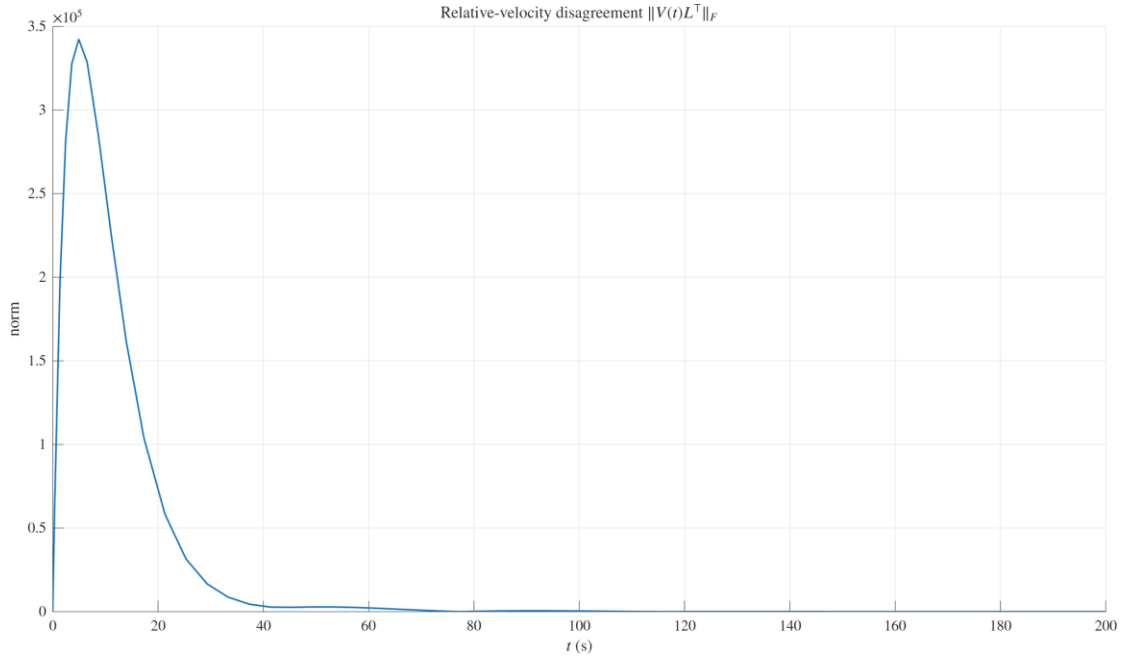
The simulated mean position of the system, defined as $\bar{R}(t) = \frac{1}{N}\sum R_i(t)$, is compared with the theoretical prediction $\bar{R}(0) + t\bar{V}(0)$.

It is clearly observed from the plots that the simulated mean position trajectory overlaps highly with the theoretical straight line in all three spatial directions. Specifically:

For the first coordinate, the mean position remains constant, indicating that the initial average velocity in this direction is close to zero.

For the second and third coordinates, the mean position exhibits a clear linear change over time. This implies that the initial average velocity is non-zero, and the group performs a translational motion at a constant velocity.

This result demonstrates that under relative damping control, the average velocity is not forced to decay to zero. Instead, the global motion mode of the group is preserved, and the evolution of the mean position is determined solely by the initial average velocity.



The evolution of the relative-velocity disagreement, defined as $\| V(t)L^T \|_F$, is shown in the final figure.

It is observed that this index exhibits a short peak during the initial phase, followed by a rapid decay, eventually approaching zero. This indicates two key points: The relative velocity differences among the agents are effectively suppressed. The system finally reaches a velocity consensus state, meaning all agents converge to the same constant velocity vector $v^* = \bar{V}(0)$.

By combining the results from the position trajectory and the disagreement plot, it is concluded that relative damping acts only on the relative motion and does not affect the global velocity component. Therefore, reaching velocity consensus in this case does not imply stopping; rather, it implies that the group moves together as a whole at a unified velocity.

2.3 Complete dynamics

1) Uncontrolled complete dynamics ($F_i = 0$)

In this section, the natural behavior of the system in the absence of any control action is investigated. When the control input is set to zero ($F_i = 0$), the dynamics of the i-th agent are determined solely by the mutual gravitational coupling. The equation of motion is expressed as:

$$m_i \ddot{R}_i = \sum_{j \neq i} \frac{G m_i m_j}{\| R_i - R_j \|^3} (R_j - R_i),$$

The gravitational interaction is reformulated using a Laplacian structure. State dependent weights are defined as:

$$w_{ij}(R) = \begin{cases} \dfrac{G m_i m_j}{\| R_i - R_j \|^3}, & i \neq j, \\ 0, & i = j. \end{cases}$$

Using these weights, a state-dependent Laplacian matrix $L(R) \in \mathbb{R}^{N \times N}$ is constructed. Consequently, the global dynamics of the system can be written in the compact matrix form:

$$M\ddot{R} = -(L(R) \otimes I_3)R,$$

where $M = \mathrm{diag}(m_1, \ldots, m_N)$is the mass matrix.

Based on the definition of the weights, the matrix $L(R)$ retains key properties of a standard Laplacian: it is symmetric, positive semi-definite, and satisfies the condition

$$L(R)\mathbf{1} = 0.$$

This implies that the system remains invariant under translational shifts.

However, a critical distinction from the linear consensus systems discussed in class is that $L(R)$is explicitly dependent on the relative positions of the agents. This makes the system a nonlinear and state-dependent Laplacian system. Furthermore, as the distance between agents approaches zero ($\| R_i - R_j \| \to 0$), the weights may become unbounded, introducing potential singularities.

$$L(R) = \begin{bmatrix} w_{12}+w_{13}+w_{14}+w_{15} & -w_{12} & -w_{13} & -w_{14} & -w_{15} \\ -w_{21} & w_{21}+w_{23}+w_{24}+w_{25} & -w_{23} & -w_{24} & -w_{25} \\ -w_{31} & -w_{32} & w_{31}+w_{32}+w_{34}+w_{35} & -w_{34} & -w_{35} \\ -w_{41} & -w_{42} & -w_{43} & w_{41}+w_{42}+w_{43}+w_{45} & -w_{45} \\ -w_{51} & -w_{52} & -w_{53} & -w_{54} & w_{51}+w_{52}+w_{53}+w_{54} \end{bmatrix}.$$

In the unforced case ($F_i = 0$), the system represents a nonlinear conservative system dominated by gravitational potential energy. It does not possess inherent damping or artificial consensus mechanisms. Therefore, the standard convergence results for linear, time-invariant Laplacians do not directly apply.

It is concluded that without control, velocity or position consensus cannot be guaranteed. The motion of the agents is highly dependent on initial conditions and may result in oscillations, clustering, or collisions. This analysis demonstrates that natural gravitational coupling alone is insufficient for coordination, thereby motivating the design of the distributed control laws introduced in the subsequent sections.

2) Distributed consensus control without compensation of nonlinear couplings

As discussed in the previous section, when the control input is zero ($F_i = 0$), the system is driven solely by nonlinear gravitational coupling. Since this system is conservative and lacks inherent damping, velocity or position consensus cannot be guaranteed.

The objective of this section is to design a distributed control law to ensure that the multi-agent system achieves consensus behavior. A critical requirement is that this must be done without compensating for or canceling the existing nonlinear gravitational coupling. In other words, the gravitational term is treated as an unremovable internal coupling, and the control law acts as an additional distributed mechanism.

Based on the original dynamic model:

$$m_i\ddot{R}_i = \sum_{j\neq i} \frac{Gm_im_j}{\| R_i - R_j \|^3}(R_j - R_i) + F_i,$$

the following distributed consensus control law is introduced:

$$F_i = -K_i \sum_{j\in\mathcal{N}_i} w_{ij}\,(R_i - R_j) - \gamma K_i \sum_{j\in\mathcal{N}_i} w_{ij}\,(\dot{R}_i - \dot{R}_j).$$

To eliminate the influence of mass differences on the acceleration level, the gain is selected as:

$$K_i = m_iK,$$

where $K > 0$ is a unified scalar gain. Consequently, the closed-loop acceleration equation is obtained as:

$$\ddot{R}_i = \sum_{j\neq i} \frac{Gm_j}{\| R_i - R_j \|^3}(R_j - R_i) - K \sum_{j\in\mathcal{N}_i} a_{ij}\,(R_i - R_j) - \gamma K \sum_{j\in\mathcal{N}_i} a_{ij}\,(\dot{R}_i - \dot{R}_j).$$

It is important to note that this control law **does not contain any model information** regarding the gravitational force (such as G or mass m_j). Therefore, it strictly satisfies the requirement of without compensating for the existing nonlinear couplings.

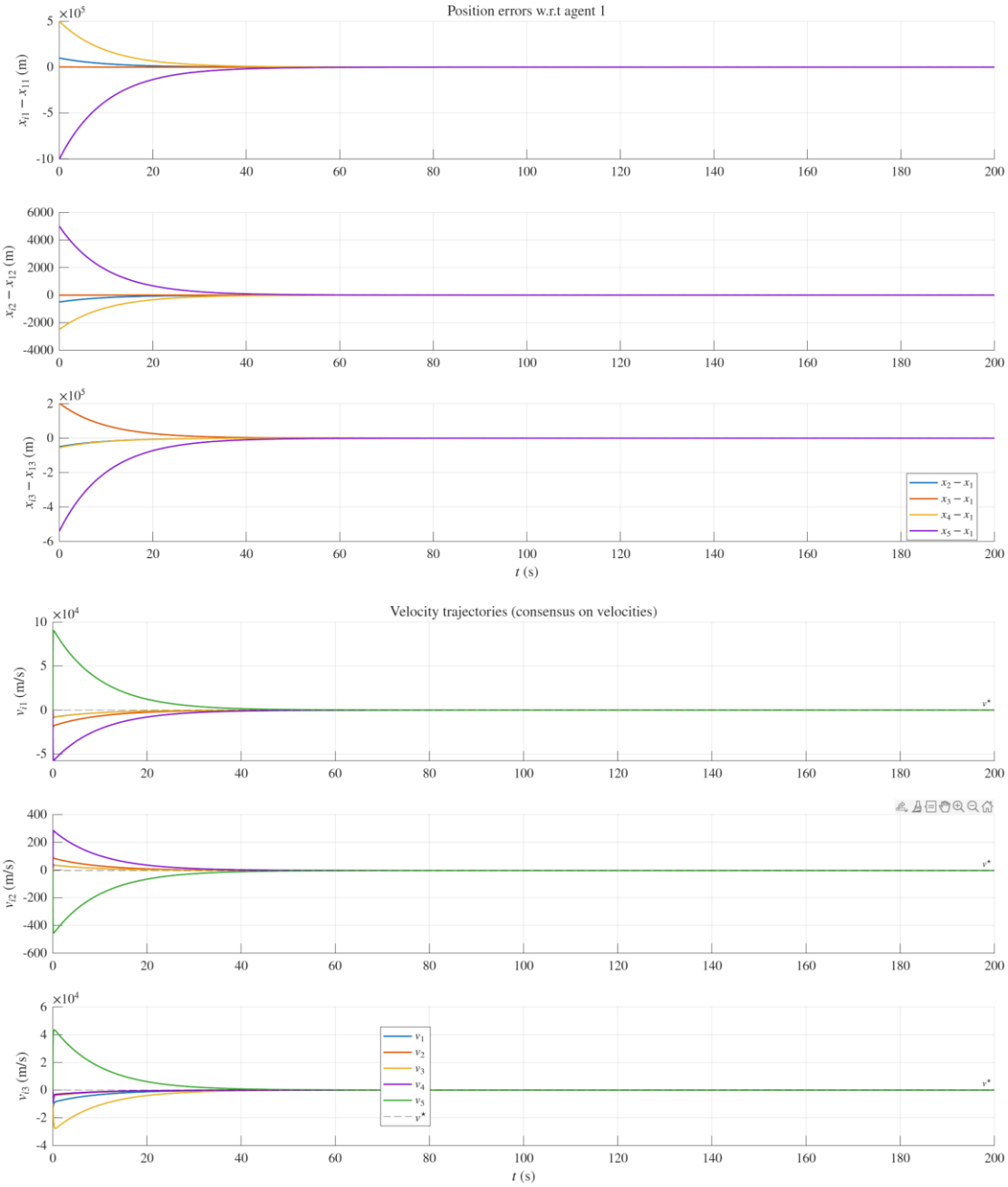
The simulation results demonstrate that the system exhibits clear consensus behavior under the designed control law.

Velocity Consensus: From the velocity trajectories, it is observed that although large differences exist initially, the velocities of all agents eventually converge to the same constant vector. This indicates that velocity consensus is achieved. Significant transient changes may occur in the early phase due to the combined effect of second-order dynamics and nonlinear gravity.

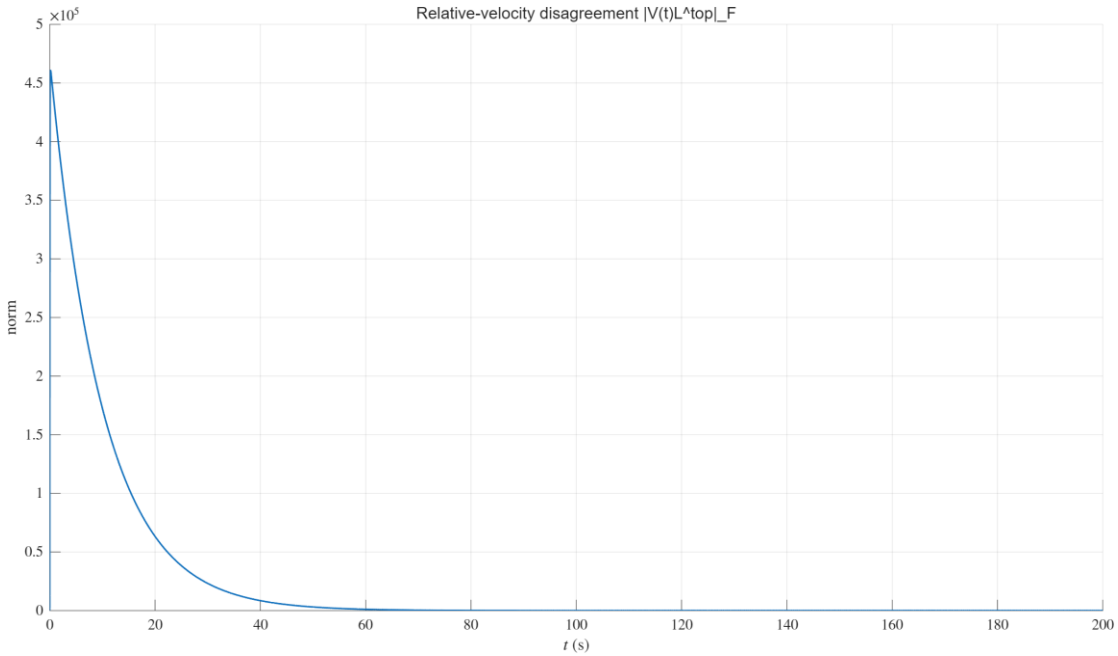
Disagreement Analysis: The relative-velocity disagreement index $\eta(t) = \| V(t)L^T \|_F$ is observed to decay monotonically and finally approach zero. This result provides quantitative verification that the relative velocity differences among agents are eliminated.

Position Synchronization: Furthermore, the relative position errors also decay over time. This implies that after velocity consensus is established, the relative positions of the system also synchronize. Although the gravitational term introduces additional nonlinear effects during the

transient phase, it does not destroy the final convergence of the system.



In summary, by introducing a distributed consensus controller based on a fixed topology, the system achieves velocity and position consensus without compensating for the nonlinear gravitational coupling. The results verify the robustness of the second-order consensus theory and its applicability in complex dynamic environments.



Conclusion

In this project, we analyzed the coordination control of a 5-body multi-agent system. Through theoretical analysis and simulation, we verified how different control laws and constraints affect the consensus. The main conclusions are:

First-Order Dynamics and Delays:

We confirmed that the standard consensus algorithm works well for undirected graphs. The convergence speed depends on the control gain K. However, the system is sensitive to time delays. While measurement delays only cause oscillations, input delays are much more critical. If the input delay exceeds the limit, the system becomes unstable and the velocities diverge.

Second-Order Dynamics:

We designed control laws proportional to the mass of each agent ($K_i = m_i K$) to handle different masses. We compared two damping strategies:

Absolute Damping:This forces the system to stop. Both velocity and position errors decay to zero, and the agents converge to a fixed point.

Relative Damping:This allows the group to keep moving. The agents reach the same velocity consensus and move together as a flock.

Complete dynamics:

In the final part, we added nonlinear gravitational forces. The simulation results show that our distributed control law is robust. Even without canceling the gravity term, the system still achieves velocity consensus and position synchronization.

In summary, this report demonstrates that Laplacian based control is effective for coordinating multi-agent systems. It works well even in complex environments with gravity, provided that communication delays are kept within a safe range.